Numerical processing of financial data

Option hedging: dynamics and simulation

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In this session

- A dynamical model for the underlying asset: the log-normal model, and its Monte Carlo simulation.
- Simulation of the Black-Scholes option hedging.
- Numerical verification of the consequences of the theta-gamma PnL approximation.

Option hedging

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  whose aim is to target the final value $g(S_T)$ of the option payoff

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  where V_t^0 is the running value of the cash position and $\delta_t \, S_t$ the value of the delta position.
- The initial portfolio value V_0 is given by the initial option price P_0.

Option hedging, some notes

- The option pricing/hedging problem
  - is typically formulated in **continuous time**
  - In this case, the mathematical formalism requires continuous-time stochastic processes (such as Brownian motion).
- The continuous-time setting:
  - a theoretical framework where the hedging portfolio can be rebalanced continuously in time
  - mathematically, this means that the time-step Δt between rebalancing dates goes to zero.

Option hedging, some notes

- In the continuous time setting
  - it can be shown that the Black-Scholes solution (that is, price function $P(t, S)$ given by the Black-Scholes PDE and delta function $\delta_t = \partial_S P(t, S_t)$) provides a *perfect hedging*.
  - Perfect hedging means

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- In our derivation of the BS solution (lecture 2), we have simplified the formalism considering a **discrete-time setting**, leading to Taylor approximations. This is why we only get

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- Terminology: the difference $V_T - g(S_T)$ (which is the option seller PnL at maturity) is also called the *hedging error*.

Option hedging

- Note that, more in general, we can look for a solution of the hedging problem using a choice of **parametric functions**

$$P_\theta(t, S_t) \qquad \delta_\theta(t, S_t).$$

- In the Black-Scholes solution:
  - the price function $P_\theta(t, S_t)$
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  depend on a single **one-dimensional parameter** $\theta = \sigma$.

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- Other solutions are possible (related for example to *stochastic volatility models*).

- We could also leverage on *deep learning* to tackle this problem

A model for asset dynamics

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A model for asset dynamics

- We can test the performances of a given delta-hedging strategy over simulated paths of the underlying asset S.
- To do so, we need a *dynamic model* for the asset price S that we can simulate over a sequence of observation times $(t_i)_{0 \leq i \leq N}$.
  - equally spaced : $t_{i+1} - t_i = \Delta t$ for all i (but we could easily relax this condition).
  - we can think that t_i is the market close of date i.

A model for asset dynamics

- We could consider a model with **additive random shocks** (also called additive noise)

$$\begin{aligned} S_{t_{i+1}} &= S_{t_i} + \text{ random perturbation} \\ &= S_{t_i} + \mu\,\Delta t + \sigma\,\sqrt{\Delta t}\,G_i\,, \end{aligned}$$

  where

  - μ is a drift parameter (dimension : $\frac{1}{\text{time}}$).
  - σ is a volatility parameter (dimension : $\frac{1}{\sqrt{\text{time}}}$).
  - $(G_i)_{i\geq 1}$ is a sequence of independent random variables with Gaussian distribution $\mathcal{N}(0,1)$.

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  - $(G_i)_{i\geq 1}$ is a sequence of independent random variables with Gaussian distribution $\mathcal{N}(0,1)$.

- Problem : the positivity of the trajectory of the asset price S is not granted, because the G_i can take arbitrarily large negative values.

A model for asset dynamics

- Instead, we rather consider a **multiplicative model**

$$\begin{aligned} S_{t_{i+1}} &= S_{t_i} \times \text{ positive perturbation} \\ &= S_{t_i}\, e^{\mu \Delta t + \sigma \sqrt{\Delta t}\, G_i} \end{aligned} \tag{1}$$

  where μ is still identified as a drift parameter, and σ as a volatility parameter.

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- Note that, using the Taylor expansion of the exp function for small time intervals Δt and for small shocks $\sqrt{\Delta t}\, G$, the model above is close to

$$S_{t_{i+1}} \approx S_{t_i} \big(1 + \mu \Delta t + \sigma \sqrt{\Delta t}\, G_i + \frac{1}{2} \sigma^2 \Delta t\, G_i\big), \tag{2}$$

  keeping terms up to order one in Δt.

  - The dynamics in eq (2) is an approximation of the dynamics in eq (1).
  - Nevertheless, the right hand side in (2) is not granted to remain positive, while the right hand side in (1) is.

Asset realized variance

A link with the sessions by M. Garcin:

- The realized variance of S over n periods $t_0, ..., t_n$ is defined from log-returns by

$$\tilde{\sigma}_n^2 := \frac{1}{n}\sum_{i=0}^{n-1}\left(\log\frac{S_{t_{i+1}}}{S_{t_i}}\right)^2 - \left(\frac{1}{n}\sum_{i=0}^{n-1}\log\frac{S_{t_{i+1}}}{S_{t_i}}\right)^2$$

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- If $\Delta t = 1$ day, the **annualized realized variance** is then given by

$$\sigma_n^2 = 250 \times \tilde{\sigma}_n^2$$

  where 250 is a (conventional) number representing the average number of market business days in one year (since $250 = 365$ minus weekends minus holidays).

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- In its turn, the **annualized realized volatility** is

$$\sigma_n = \sqrt{\sigma_n^2} = \sqrt{250 \times \tilde{\sigma}_n^2}.$$

Asset realized variance in the log-normal model

Small exercise

1. Write the realized variance $\tilde{\sigma}_n^2$ of log-returns in the log-normal dynamical model, where
$$\log \frac{S_{t_{i+1}}}{S_{t_i}} = \mu \Delta t + \sigma \sqrt{\Delta t}\, G_i$$
and $\Delta t = 1$ day.

2. Evaluate the limit of $\tilde{\sigma}_n^2$ as $n \to \infty$.
Conclude on the value the annualized realized variance of the asset price in the log-normal model.

Monte Carlo simulation of the hedging strategy

- In the notebook

  hedging_simulation.ipynb

  we are going to simulate

  - The trajectories of the underlying asset S in the log-normal dynamical model
  - Then, the performances of the Black-Scholes delta-hedging strategy applied along the trajectories of the asset price.